stion 1:

The Figure 1 below shows the construction of a 7490 counter IC. It has four flip flops to be used as independent mod 2 (FFA) and mod 5(FFB, FFC, FFD) counters. Application of logic 1 to the reset inputs R1 and R2 will make all the FFs to be at logic 0 whereas Logic 1 at pre-set inputs S1 and S2 will set the counter to $(1001)_2$.

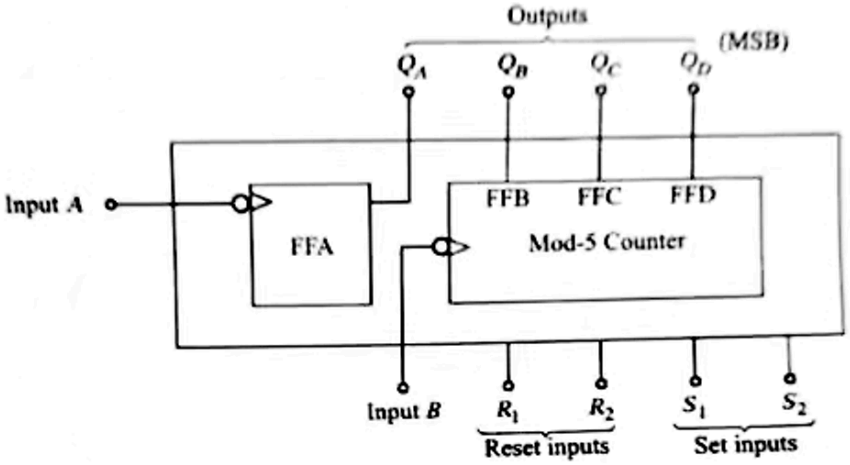


Fig 1: 7490 IC architecture

Now if the counter terminals are connected as shown in Figure 2, explain the operation of the counter and construct the counting sequence table. Consider all flip-flops are initially 0. **[3+3=6 marks]**

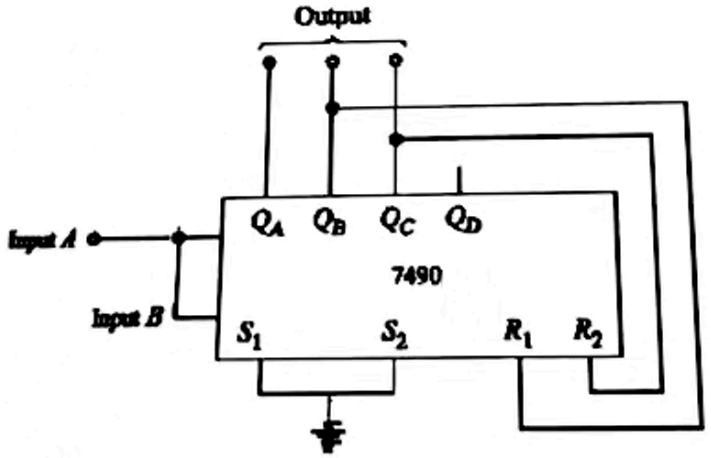


Fig 2: counter design using 7490 IC

Question 2

A shop owner wants to see how many customers are there inside the shop with a maximum capacity of 100. Can you help to design an optimized mod-100 asynchronous counter adopting only "D" type flip flops and other logic gates that will show the number of customers in two 7-segment displays? Estimate the price of the gates and flip-flops needed for the design assuming that all component prices are 2 RMB each.

[6+3=9 marks]